Vishwanth Marri

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Profile Summary

Full-stack Software Engineer with 3+ years of experience building scalable microservices, event-driven pipelines, and React/TypeScript applications at Salesforce and Virtusa. Strong in Python, Java, Node.js, Kafka, and AWS, with a focus on reliability and performance: cut API response times by 73%, maintained 99.9% uptime under peak traffic, and reduced frontend load times by 40%. Increasingly building GenAI features with the Claude API and RAG.

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, C++, Go, SQL

Frontend Engineering: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend Engineering: FastAPI, Spring Boot, Node.js, Express.js, Microservices, Kafka, RESTful APIs, Webhooks

AI & LLMs: Claude API, LangChain, LangGraph, Model Context Protocol (MCP), RAG, AI Agents, Prompt Engineering, LLM Evaluation, Structured Outputs

Databases: PostgreSQL, pgvector, MySQL, MongoDB, DynamoDB, Elasticsearch, Redis

Cloud & DevOps: AWS (ECS, EC2, S3, RDS, Lambda, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Testing, Security & Tools: Pytest, Jest, OAuth 2.0, JWT, RBAC, Structured Logging, Git, Claude Code, Cursor

Professional Experience

Software Engineer, *Salesforce* 01/2026 – Present | Dallas, TX

- Developed CRM microservices and Kafka-based event pipelines with bi-directional webhooks in Python, using AWS Lambda for event processing and reducing cross-system sync errors by 90% across enterprise platforms.
- Built a Generative AI pipeline using FastAPI and the Claude API with structured outputs to classify customer onboarding data, storing classification states in DynamoDB and eliminating a manual review step.
- Engineered React and TypeScript interfaces using lazy loading, code splitting, and reusable UI components to optimize bundle sizes and reduce frontend load times by 40%.
- Enforced least-privilege access across 15+ microservices and 4 user roles using RBAC and JWT, with CloudWatch alerts that caught access errors within minutes, before customers were affected.

Software Engineer Intern, *Kontaktsource* 08/2025 – 12/2025 | Miami, FL

- Developed internal data-validation pipelines using Node.js and FastAPI to automatically scrub and validate member application data prior to manual review, processing 5k+ weekly entries and reducing false flags from operations by 20%.
- Built modular React components and RESTful API endpoints for core membership, application, and event portals, standardizing forms and improving WCAG accessibility compliance.
- Resolved recurring form-submission defects and engineered webhook payload validation scripts for CRM integrations, documenting debugging guides that reduced repeat support tickets by 30%.

Full Stack Developer, *Virtusa* 06/2022 – 07/2024 | Hyderabad, India

- Enhanced a multi-tenant client operations platform using React, Node.js, Express, and MongoDB, centralizing enterprise workflows, isolating tenant data, and automating manual processing steps.
- Engineered fault-tolerant backend services using Java (Spring Boot) and Node.js with asynchronous processing and structured retries, maintaining 99.9% uptime and keeping third-party enterprise API integrations stable under peak traffic.
- Containerized microservices using Docker and orchestrated deployments on AWS ECS via GitHub Actions CI/CD pipelines, accelerating software release cycles by 35%.
- Optimized PostgreSQL performance using targeted indexing, query tuning, and Redis caching, cutting average API response times from 450ms to 120ms for high-volume workflows.
- Built unit and integration test suites across APIs, UI components, and data layers, reaching 90%+ code coverage and catching defects before production.

Projects

AI-Powered Document Q&A System | FastAPI, pgvector, LangChain, Claude API, MCP, TypeScript, React

- Built and deployed a production document Q&A platform featuring two-stage retrieval (pgvector ANN search + Voyage AI reranking), LangChain orchestration with MCP tool use, and real-time Claude API streaming via Server-Sent Events.
- Resolved a concurrent race condition causing silent data corruption by implementing an idempotent async job queue, maintaining sub-200ms retrieval latency under concurrent load.

Lead Enrichment & Outreach Agent | Python, FastAPI, LangGraph, Claude API, Structured Outputs, Pytest

- Built a multi-step LangGraph agent that enriches company data from public sources, qualifies leads against configurable criteria, and drafts personalized outreach using Claude API structured outputs, validated against live company data.
- Designed an independent grading guardrail that verifies every drafted claim against its source before human review and fails closed on verification errors instead of auto-approving unverified output, covered by integration tests including concurrent multi-session runs.

Education

University of Cincinnati Cincinnati, OH

Master of Engineering in Computer Science 08/2024 – 05/2026